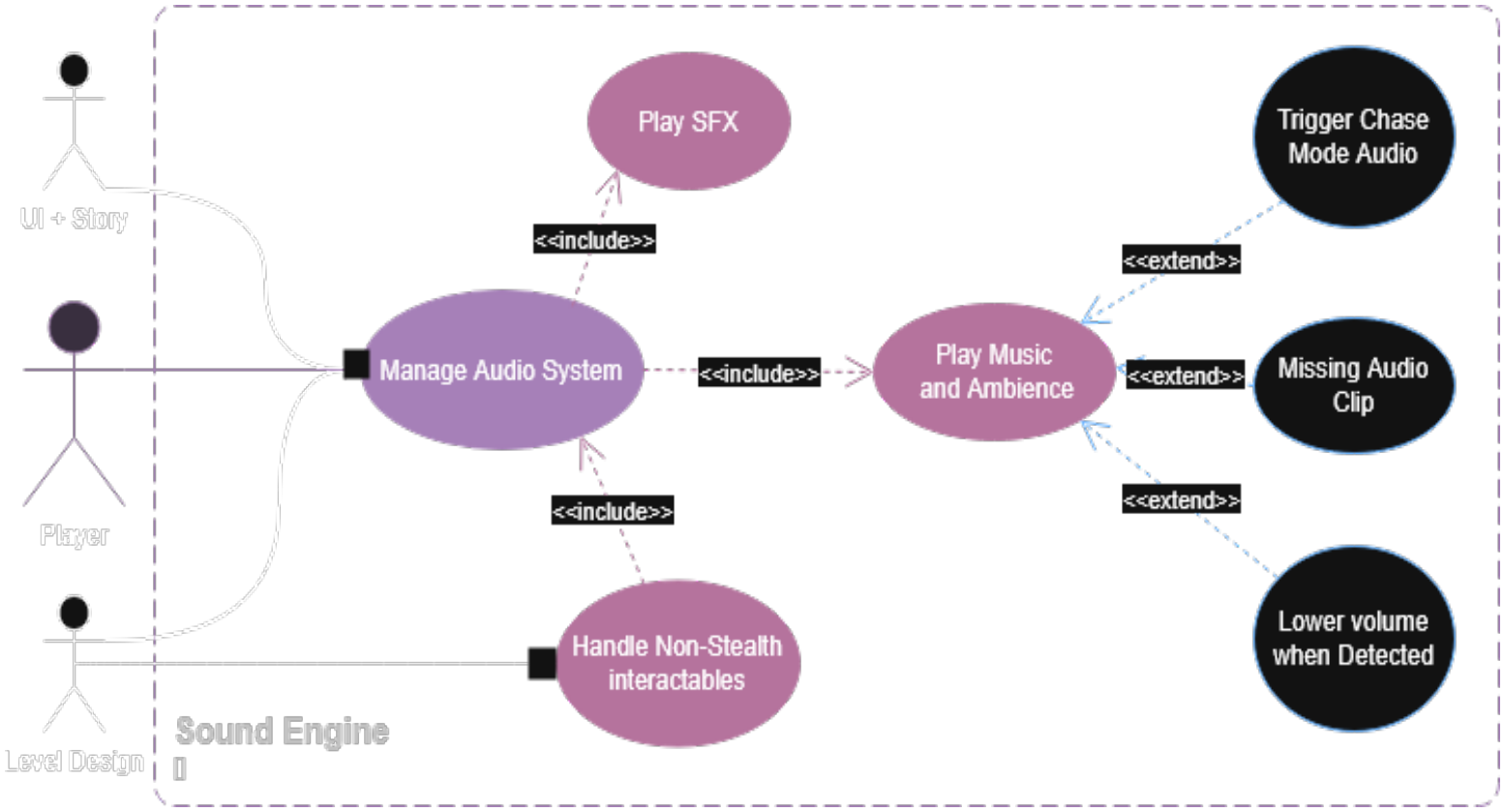
Name _____Nastia Kossiak_____ Mark_____ /50

1. Brief introduction_____ /3

This document describes my design and implementation of the Sound System and Non-Stealth Interactables for “Fruit of the Witch”. My features will include overall background music and ambience, global sound effects, volume adaptation, and non-stealth objects/assets for the player to interact with. The goal of these implementations is to ensure that the player has the proper auditory feedback revolving around their stealth decisions and adding into the gameplay progression by making the game as immersive as possible.

2. Use case diagram with scenario _____/14]

Use Case Diagram



Scenario

Name: Manage Audio System

Summary:

Player starts the game, initializing the Manage Audio System, which handles all audio throughout the game. The system plays UI and player SFX, triggers SFX from non-stealth interactables placed by the Level Design subsystem, and manages both menu music and level music/ambience during gameplay.

Actors:

Player, Level Design, UI+Story.

Preconditions:

The game application should be in execution and all audio assets loaded. The UI and level design subsystems are active and able to send requests to the Sound Engine subsystem.

Basic sequence:

Step 1: Player starts game and boots into Main Menu, where Audio System loads the Menu Music and the UI SFX prefab.

Step 2: Player presses Start and the corresponding level music & ambience starts playing.

Step 3: The player moves and plays the main logic of the game where the corresponding sound effects play; the player successfully traverses through all the levels.

Exceptions:

2. The corresponding audio file is missing, so the Audio System plays a default track.

3.1. The player triggers enemy detection, which will cause a special sound effect to be played and the global audio to lower.

3.2. The player is not able to hide from the enemy, so a special chase soundtrack will play while the player is being chased until they manage to hide again.

Postconditions:

Manage Audio System remains active and continues to handle the corresponding audio requests and exceptions. When the player completes the game, the Audio track will play the menu music again.

Priority: Medium-High

ID: AUDIO-01

3. Data Flowdiagram(s) from Level 0 to process description for your feature _____ 14

Context Diagram

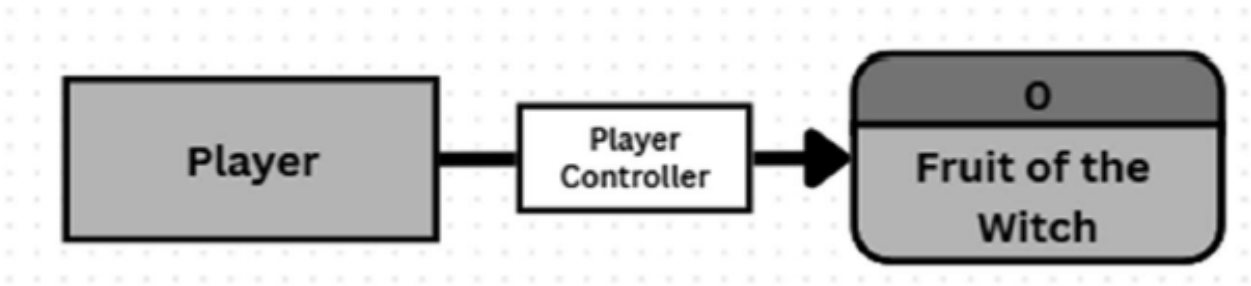
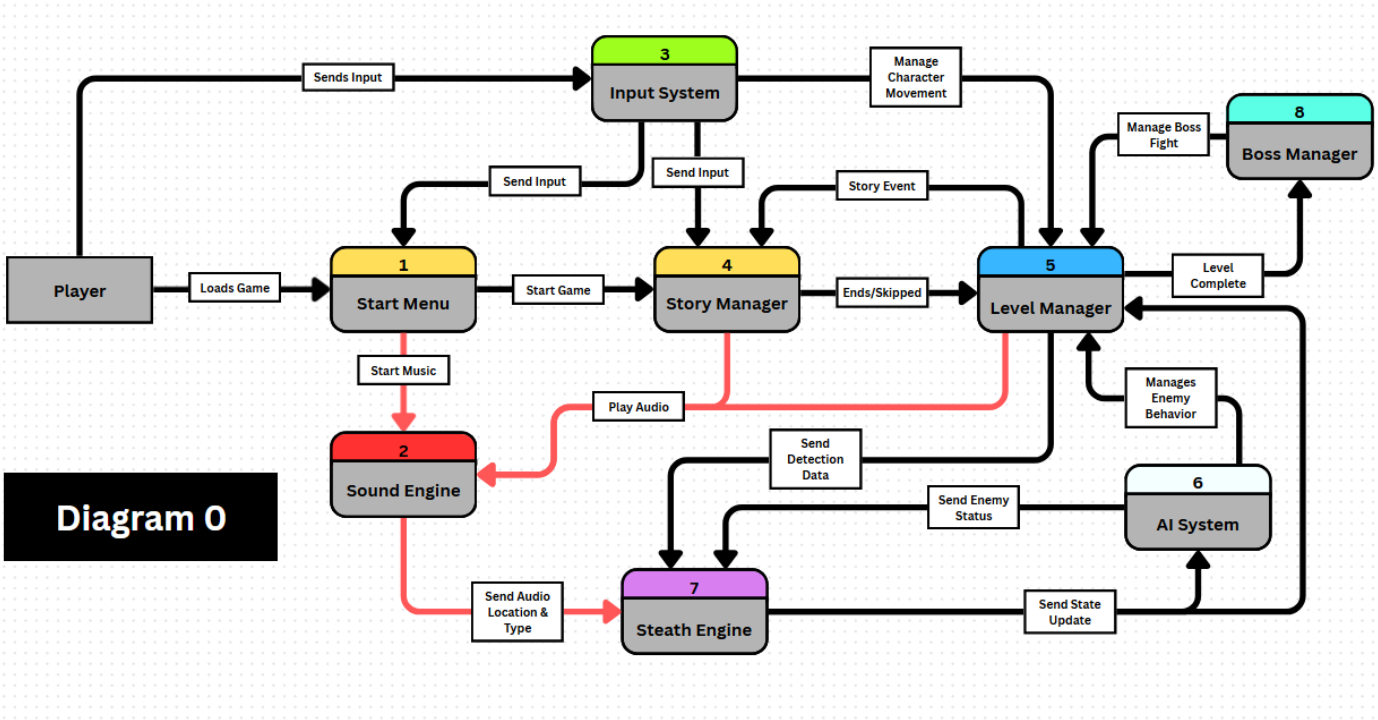
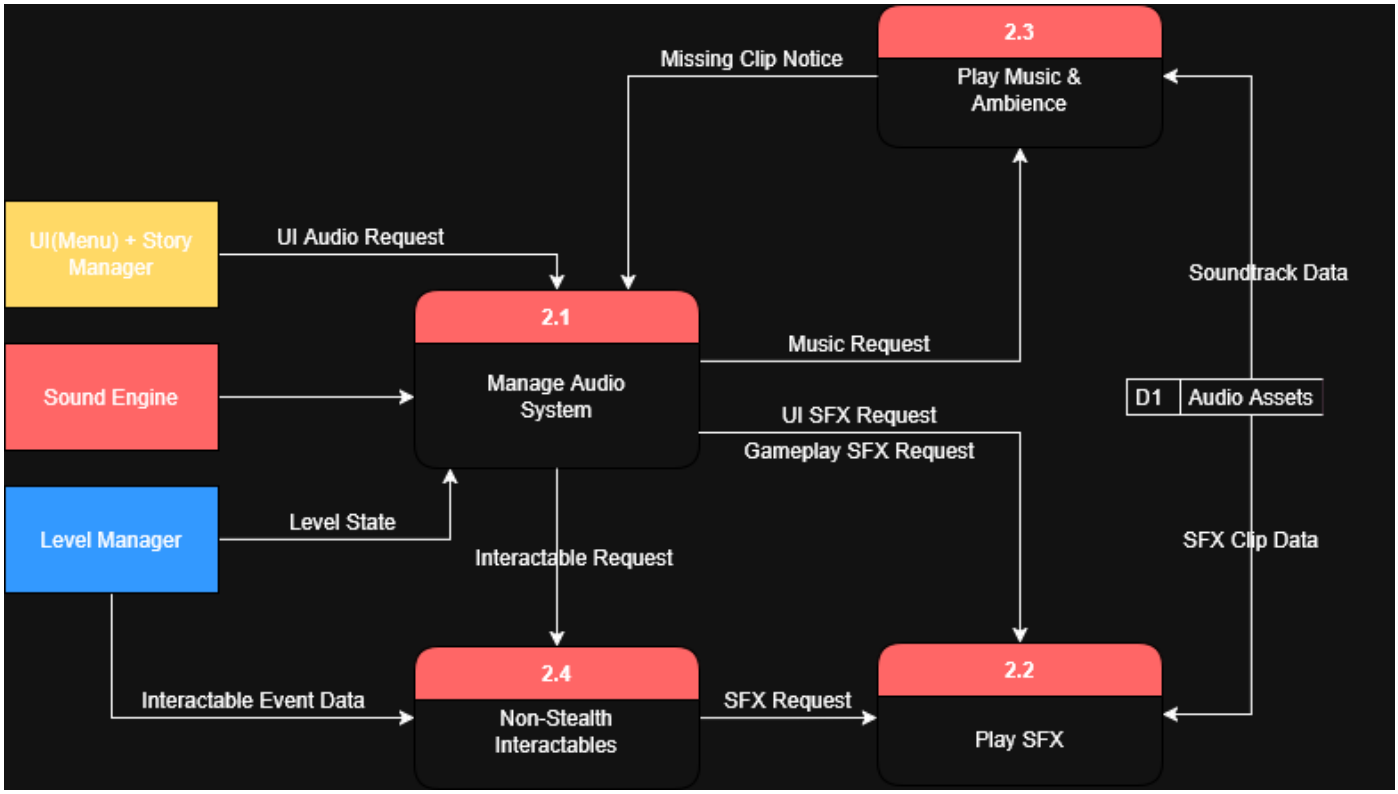


Diagram 0



Level 1 DFD:



Process Descriptions

Manage Audio System

Pull UI Audio Request, Level State Data, and Interactable Event Data.
IF request type is UI audio THEN
Send SFX Request to Play UI SFX.
END IF
ELSE IF request type is gameplay SFX THEN
Send SFX Request to Play Gameplay SFX.
END IF
ELSE IF request type is music request THEN
Send Music Request to Play Music & Ambience.
END IF
ELSE IF request type is interactable event THEN
Send Interactable Request to Handle Non-Stealth Interactables.
END IF

Play SFX

Pull SFX Request.
Retrieve SFX Clip Data from Audio Assets.
IF SFX clip exists THEN

Play requested SFX.
END IF
ELSE
Send Missing Clip Notice to Manage Audio System.
Play default SFX.
END IF

Play Music & Ambience

Pull Music Request and Level State Data.
Retrieve Soundtrack Data from Audio Assets.
IF player state is detected THEN
Lower global volume.
Play detection SFX.
END IF
ELSE IF player state is chased THEN
Play chase soundtrack.
END IF
ELSE IF soundtrack exists THEN
Play requested music or ambience.
END IF
ELSE
Send Missing Track Notice to Manage Audio System.
Play default soundtrack.
END IF

Handle Non-Stealth Interactables

Pull Interactable Event Data.
Identify interactable type and location.
Generate Interactable SFX Request.
Send Interactable SFX Request to Play SFX.

4. Acceptance Tests _____9

Run feature 1000 times sending output to a file.

The output file will have the following characteristics:

- UI interactions always have the correct hover or click sound.
- Gameplay actions always play the corresponding gameplay SFX.
- Interactable events always have the correct interactable SFX.
- Menu music always starts when the game enters the main menu and cutscenes.
- Level music and ambience always start when a level begins.
- Detection state lowers the global audio volume appropriately.
- Chase state replaces the normal level music with the chase soundtrack.
- If an audio asset is missing, the system plays a default fallback sound.
- The audio system continues functioning throughout gameplay.
- The audio system sends and receives the correct signals from the corresponding subsystems.

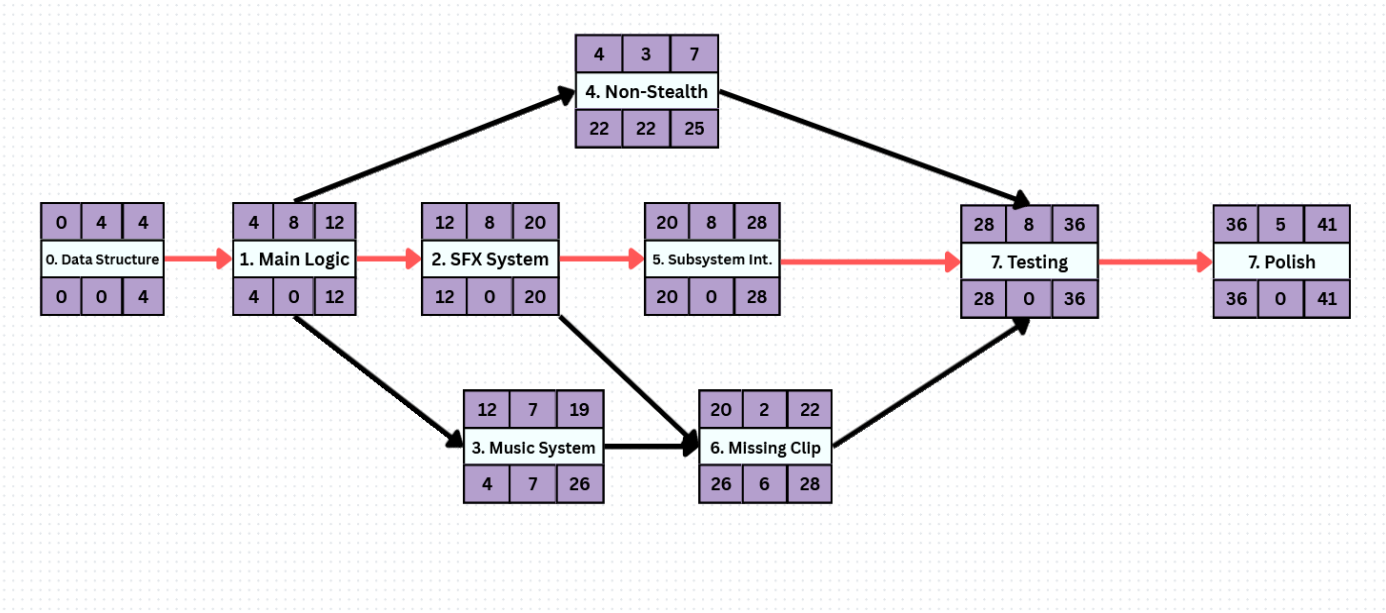
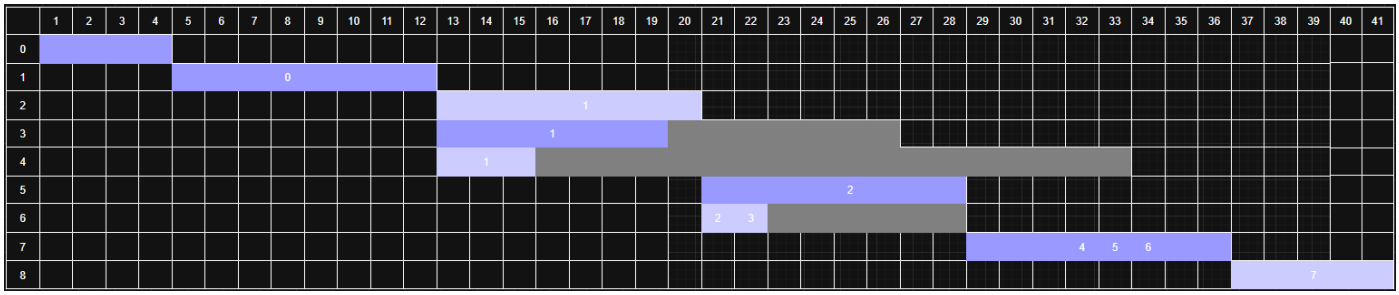
Input	Source	Output	Notes
Click Event	UI+Story	UI SFX plays	Confirms UI Audio routing
Gameplay event	Player and Level	Gameplay SFX plays	Verify that the audio system plays SFX
Interactable activation	Level Manager	Interactable SFX plays	tests handle Non-stealth interactables
Level Transition	Level Manager	Level soundtrack plays	Confirms if it is the correct level music selection
Player detected by enemy	Stealth Engine	Global volume lowers	Testing exception case for detection audio
Chase Mode	Gameplay state change	Chase special soundtrack plays	Confirms if the special chase music overrides the current level soundtrack and then goes back to play the level music when the player hides.
Missing Audio Clips	Audio Assets lookup	Default SFX plays	Boundary case

5. Timeline _____ /10

Work items

Task	Duration (PWks)	Predecessor Task(s)
0. Asset and Data Structure of assets	4	-
1. Manage Audio System Main Logic	8	0
2. SFX System Implementation	8	1
3. Music & Ambience Implementation	7	1
4. Non-Stealth Interactable Audio	3	1
5. Subsystem Integration	8	2
6. Missing Clip Handling	2	2,3
7. Testing	8	4, 5, 6
8. Polish	5	7

Pert diagram



Gantt timeline